CLASS 7 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. In the reaction above, the colour of the solution changes from: 1

(a) blue to colourless **(b)** green to blue
(c) blue to green **(d)** colourless to blue

2. A student heats a piece of wax and sees it melt. On stronger heating, black smoke appears. The black smoke indicates: 1

(a) no change **(b)** evaporation only
(c) melting only **(d)** unburnt carbon, a chemical change

3. A tablet that fizzes in water is likely to contain: 1

(a) a carbonate and an acid **(b)** only sugar
(c) only salt **(d)** only a base

4. A tester glows dimly with a pencil lead and brightly with a copper wire. This shows that: 1

(a) both conduct, copper better **(b)** only graphite conducts
(c) neither conducts **(d)** copper is an insulator

5. A student has four metals P, Q, R, S. P and Q react with dilute HCl, R and S do not. Q reacts faster than P. The most reactive is: 1

(a) P **(b)** Q **(c)** S **(d)** R

6. Why are bicycle chains regularly oiled? 1

(a) to make them brittle **(b)** to make them heavier
(c) to prevent rusting and reduce friction **(d)** to make them shiny only

7. A student adds drops of base to acid and notes the litmus. The sequence of colours is: 1

(a) red, purple, blue **(b)** blue, yellow, red
(c) red, green, blue **(d)** blue, purple, red

8. When a student removes one bulb from a string of decorative bulbs, all the bulbs go off. This shows that the circuit has: 1

(a) no cell **(b)** no wire
(c) a single path for the current **(d)** many separate paths

9. A metal M does not displace copper from CuSO4 but copper displaces it from M's salt solution. M is: 1

(a) as reactive as copper **(b)** less reactive than copper
(c) more reactive than copper **(d)** a non-metal

10. A student says that a burning candle makes the wax vanish. The correct statement is: 1

(a) the wax turns into the wick
(b) the wax only melts
(c) the wax changes into new substances such as carbon dioxide and water vapour
(d) the wax is destroyed

11. A student mixes equal volumes of acid and base. The mixture still turns blue litmus red. This suggests: 1

(a) the acid was stronger or more concentrated
(b) the mixture is neutral
(c) the base was stronger
(d) no reaction

12. If the bulb in the tester does not glow with iron nail, it could be because: 1

(a) nail is a plastic **(b)** iron is an insulator
(c) the nail is coated with paint or rust **(d)** cell is stronger

13. **Assertion (A):** Graphite conducts electricity even though it is a non-metal. 1
Reason (R): Graphite is a form of carbon.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** Rusting of iron takes place only when both oxygen and water are present. 1
Reason (R): Iron is a metal.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** Neutralisation is the reaction between an acid and a base to form salt and water. 1
Reason (R): Heat is released during a neutralisation reaction.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do metals feel cold to the touch? 2

17. Why is a slow chemical change not always noticed? 2

18. Why does the stomach not get damaged by its own hydrochloric acid? 2

19. Why is the heating element usually a coil of nichrome and not of copper? 2

20. Explain why a gold ring does not need cleaning as often as a silver ring. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Describe a test to distinguish a piece of copper from a piece of aluminium without chemicals. 3

22. Calcium oxide reacts with water and the mixture becomes hot. Is it physical or chemical? Give two reasons. 3

23. Explain why a student's mixture of acid and base became warm. 3

24. Explain why a tungsten filament bulb glows white hot but an iron wire would not last. 3

OR

Why is a small piece of sodium kept in kerosene reduced in size when cut? 3

25. Explain with an example how a physical change may accompany a chemical change. 3

OR

Explain how red cabbage juice could be used to compare the strength of lemon and vinegar. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Aditi helps her mother in the kitchen in Pune. The frying pan is made of aluminium but its handle is made of plastic. Aditi also notices that aluminium foil can be wrapped around parathas, and that the copper wire of the mixer is thin and long. 5

(i) Why is the pan made of a metal?

(ii) Why is the handle made of plastic?

(iii) Which property of aluminium allows it to be made into foil?

(iv) Name the property that allows copper to be made into wire and one more property that makes copper suitable for electric wires.

27. Explain, with equations, the reactions of magnesium with oxygen, acid and copper sulphate solution. 5

OR

Analyse the changes taking place when you light a gas stove under a vessel of milk, with milk boiling over and burning at the bottom. Classify each change. 5

END OF QUESTION PAPER — Total: 50 Marks